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Easily construct, covert or build Likert items from vectors or data.frame

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Executive Summary

Signatories

Project team

Contributors

Consulted

The Problem

The CRAN packages for Likert-type data analysis workflow within the R statistical computing ecosystem, viz, Mollazehi (2025); Winzar (2026) and Bryer and Speerschneider (2025), has not contemplated a situation whereby the items may be transliterated to other formats for more robust analysis and interpretation. They effectively facilitate standard Likert scale workflows, but operate under the assumption that input data is structured in the conventional, clean ordinal format. Consequently, the existing software landscape lacks the architectural framework required to handle data format transliteration.

In empirical research, investigators frequently encounter non-standard data structures that require transformation to achieve robust analysis and deep interpretation. For instance, researchers often face the need to convert Likert items text responses into ordinal Likert scales and vice versa, or synthesize valid Likert-type scales from non-Likert source data. Because current R tools cannot execute these complex transformations natively, research teams are forced to rely on disjointed, manual workarounds and ad-hoc scripting. This fragmentation severely disrupts reproducible data workflows, introduces significant risks of data entry and manipulation errors, and results in substantial losses in operational man-hours. This proposed project addresses this computational gap by proposing a novel, open-source R package framework. By proposing a bi-directional transliteration and dynamic format conversion facilities, the software will eliminate manual data manipulation, preserve data integrity, and significantly optimize workflow efficiency for researchers handling complex, non-standard behavioral and social science data.

The proposal

Overview

Detail

Minimum Viable Product

Architecture

Assumptions

External dependencies

Project plan

Start-up phase

Technical delivery

Other aspects

Budget & funding plan

Milestone	Expected Completion period	Deliverables / Tangible Work Products
Inception and take-off	Two months	Complete architecture design and initial abstract class definitions in the GitHub repository.
Coding and testing	Four months	Release the optimized core chunking engine; pass integration tests with local memory backends. Conduct a sensitization workshop for faculty and students.
Preliminary	Two months	Initial submission of version 1.0.0 to CRAN, complete release comprehensive pkgdown documentation website, and host a community webinar.
Final release and closure	One month	Final CRAN submission and project closure

Success

Definition of done

Measuring success

Future work

Bryer, Jason, and Kimberly Speerschneider. 2025. *Likert: Analysis and Visualization Likert Items*. https://doi.org/10.32614/CRAN.package.likert.

Mollazehi, Mohammad. 2025. *LikertEZ: Easy Analysis and Visualization of Likert Scale Data*. https://doi.org/10.32614/CRAN.package.LikertEZ.

Winzar, Hume. 2026. *LikertMakeR: Synthesise and Correlate Likert Scale and Rating-Scale Data Based on Summary Statistics*. https://doi.org/10.32614/CRAN.package.LikertMakeR.